

BookHaven

24.9.2026

Brief

BookHaven is a remake of my previous book management application, rebuilt using **Next.js** and **MongoDB**. The goal is to rethink and improve the existing functionality while introducing new features and a more modern application structure.

Tech

Next.js, used as the main application framework for building the frontend, handling routing, server-side functionality, and API endpoints.

Tailwind CSS, used for styling the user interface and creating a responsive layout.

MongoDB, used as the database for storing books, user information, and other application data.

JWT, used for authentication tokens.

bcrypt / bcryptjs, for hashing user passwords.

React-hot-toast, for giving user feedback as toasts.

Jest and **Playwright**, for testing code.

Models (Frontend)

Book

Field	Type	Description
title	string	Name of the book
author	string	Name of book author
coverImage	string	URL to image source
rating	float / null	Reader's rating for the book, null if not read yet but added to databse
readDate	Date / null	When book was read, null if not read yet but added to databse
pageCount	integer / Number	How many pages does the book have
id	Number	id number for book

User

Field	Type	Description
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username	string	Name for user
password	string	User's password
id	Number	Id number for user

Collections (Database)

book

Field	Type	Description
_id	ObjectId	id for the book
authorId	ObjectId	reference to author
coverImage	string	url to the book cover source
title	string	name of the book
pageCount	Number	Number of pages

author

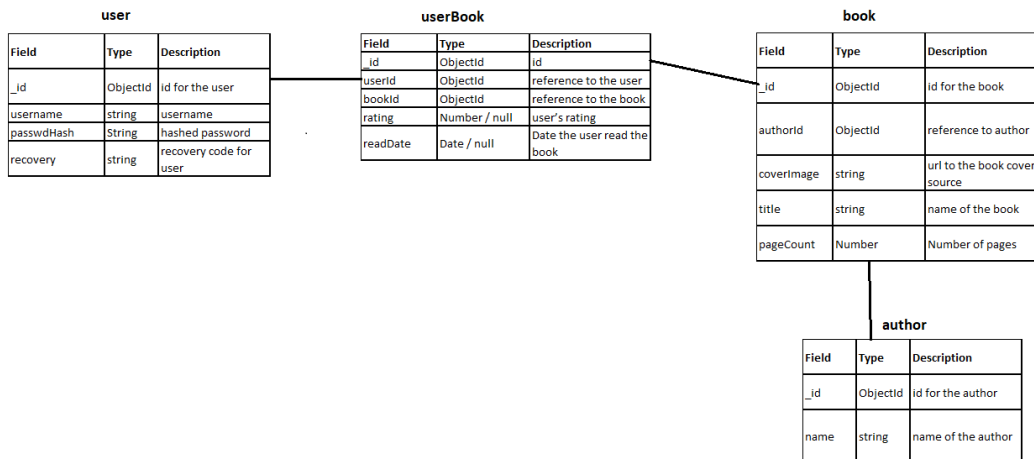
Field	Type	Description
_id	ObjectId	id for the author
name	string	name of the author

user

Field	Type	Description
_id	ObjectId	id for the user
username	string	username
passwdHash	String	hashed password
recovery	string	recovery code for user

userBook

Field	Type	Description
_id	ObjectId	id
userId	ObjectId	reference to the user
bookId	ObjectId	reference to the book
rating	Number / null	user's rating
readDate	Date / null	Date the user read the book



Views

Home

Landing page for the application.

Should display some info about the application.

Should have buttons for login and registering.

Login

Accessed from Home page.

Login page for the application.

Should have form for login credentials like username and password.

Should have feedbacks, for successful login or failure.

Should have option for staying logged in.

Register

Accessed from Home page.

Page for registering to the application.

Should have form for creating a user, username and password.

Should have feedbacks for failing to create a user, such as server error, database error, or username taken.

Should have feedback for successful creation.

Should display user the recovery code with successful creation.

Dashboard

Requires authenticating to access from Login page.

First page logged in user sees.

Should have option to log out.

Should display amount of books in the library.

Should display recently added books, max 3.

Should display recently read books, max 3. . (readDate is recent)

Should display books that you have added but haven't read yet, max 3. (readDate is null)

Should display reading statistics, such as how many books you read in last 7 days, total pages so far (all, and in a year)

Should display average rating of books.

Library

Accessed from Dashboard.

List of all the books.

Should show all the books, read and unread.

Should allow books to be searched by title or author.

Should be able to sorted by author, title, read date, or rating.

Should have tabs for different read date years.

Should open up View Book page when clicking a book.

Add book

Accessed from Dashboard, or Library.

Page where user can add a book to library.

Should have inputs for title, author, page count, rating and option for "not read yet".

Should have a input for book cover URL, but to be optional.

Would be nice to have a preview for the book cover.

Should give feedbacks to user upon success or failure to add a book.

Would be nice to retrieve the cover from online without user adding the url.

Would be nice to show already added authors, so user does not have to write, if said author is already in database.

View book

Accessed from Dashboard, or Library.

Page where user can examine added book.

Should display info about the book like title, author, read date, rating, and page count.

Should display book cover.

Should have button for deleting the book.

Should have button to open up editing.

Should have button to mark book read (IF UNREAD)

Edit book

Accessed from View Book

Page where can user can edit added book.

Should have inputs for title, author, page count, rating and option for “not read yet”.

Should have a input for book cover URL.

Would be nice to have a preview for the book cover.

Should give feedbacks to user upon success or failure to edit a book.

API

Authentication

Method	Endpoint	Purpose
POST	/api/auth/register	Creating a new user
POST	/api/auth/login	Authenticate user
POST	/api/auth/logout	Log out current user
POST	/api/auth/recover	Recover account using the recovery code

Dashboard

Method	Endpoint	Purpose
GET	/api/dashboard	Get data for dashboard

Books

Method	Endpoint	Purpose
GET	/api/books	Get all user books
POST	/api/books	Add a new book
GET	/api/books/[id]	Get a specific book
PUT	/api/books/[id]	Update a specific book
DELETE	/api/books/[id]	Delete specific book

